Left Without Being Seen: Nurse Work Environment and Timely Outcomes in New York and Illinois Emergency Departments

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Contribution to Emergency Nursing Practice

- The proportion of patients who leave the emergency department without being seen and patients' length of stay are 2 measures of timely care in the emergency department setting related to hospital crowding.
- While prior evidence demonstrates a relationship between the nurse work environment and patient outcomes across many settings, it is unknown whether these effects extend to timely patient outcomes in the emergency department.
- Better emergency nurse work environments were found to be associated with reductions in emergency department left without being seen rates and emergency department length of stay. To reduce barriers to timely care in emergency departments, hospital administrators should consider investing in better work environments through safe staffing resources, high-quality nurse management, and nurse involvement in unit affairs.

Abstract

Introduction: This study determined the relationship between the emergency nurse work environment and emergency department patient left without being seen rates and lengths of stay.

Methods: Cross-sectional analysis of 215 New York and Illinois emergency departments. The work environment (abbreviated Practice Environment Scale of the Nursing Work Index) was measured by emergency nurses in the 2021 RN4CAST-NY/IL survey and linked with outcomes from Hospital Compare. Regression models estimated the relationship between the nurse work environment and emergency department patient left without being seen rates, median length of stay (in minutes), and median behavioral health patient length of stay. Model coefficients were used to estimate expected additional care minutes gained if emergency department work environments improved.

Results: "Mixed" work environments had the longest median overall length of stay (3.4 hours) and the highest median left without being seen rates (2.2%), while "poor" work environments had the longest median length of stay for behavioral health patients (6 hours). Improving the emergency department work environment from poor to mixed (and mixed to better) was associated with a 13-minute reduction in overall length of stay ($P \leq .05$), a 33-minute reduction in behavioral health length of stay ($P \leq .01$), and a 19% reduction in left without being seen rates ($P \leq .01$). We estimated 11,824 to 41,071 additional patients could be seen in emergency departments associated with work environment improvements from "poor" to "better," depending on annual patient volumes.

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Discussion: Hospital administrators should consider investing in nurse work environments as a foundation to improve timely outcomes.

Key words: Emergency; Operations; Nursing; Workforce

Introduction

Emergency department crowding is considered a public health crisis[1] associated with higher rates of patients who leave the emergency department without being seen and extended ED lengths of stay (greater than 4 hours to discharge or admission).[2] When individuals experience delays in necessary emergency care, they are at greater risk for experiencing poorer health outcomes, including higher odds of mortality for time-sensitive conditions like acute myocardial infarction and stroke.[3,4] Among patients who leave the emergency department without being seen, studies have shown that up to 35% have acute conditions requiring urgent evaluation.[5,6] Improving ED throughput measures like left without being seen (LWBS) rates and patient length of stay (LOS) is critical to ensure high-quality patient outcomes and to reduce access barriers to acute care services such as prolonged wait times, boarding, and ambulance diversion.[7-9]

In a letter to the White House in late 2022,[10] leaders from national organizations, including the Emergency Nurses Association, the American Nurses Association, and the American College of Emergency Physicians described emergency departments in the U.S. as fraught with boarding and prolonged patient wait times that compromise patient care quality and safety in addition to clinicians' well-being. Indeed, in a recent study of nurses employed in 60 Magnet hospitals in 2021,[11,12] approximately half of emergency nurses reported high burnout (53.7%) and intent to leave their job (49.9%) in the next year, while close to a third (27.6%) reported high job dissatisfaction. Nurses are critical informants to their work environments—an important indicator of the context of care for nurses with regards to supportive leadership, staffing resource adequacy, interprofessional relationships, and nurse involvement in unit affairs (eg, ED throughput). Contemporary evidence demonstrates that the quality of nurse work environments varies significantly across hospital emergency departments[13] and that nurses working in poor ED work environments (compared to more favorable ones) have higher odds of reporting burnout, job dissatisfaction, turnover intention, and poor patient care quality care and safety. Overall, the nurse work environment is an important indicator of nurses' reports on their well-being[11,13] and the quality of patient care delivered in these fast-paced, high-acuity care settings.

An understudied systemic feature of health care organizations that could be leveraged to address timely care in the ED setting is the nurse work environment. Existing evidence demonstrates associations between the inpatient (ie, medical-surgical) nurse work environment and better patient outcomes, such as lower odds of health care–associated infections, hospital readmissions, and avoidable deaths.[14-16] As a unique care setting characterized by fast-paced workflows, uncertainty, and high patient acuity levels, few studies explore the relationship between the nurse work environment and patient care measures specific to the emergency department. While existing evidence firmly establishes that hospital crowding is predominately attributed to inefficiencies in inpatient[17,18] hospital operations, many of these inefficiencies (ie, boarding, prolonged ED wait times, patients leaving without being seen and/or against medical advice) manifest in the emergency department itself.

The objective of this study was to examine the relationship between emergency nurse work environments and hospital emergency department rates of patients who left without being seen and ED lengths of stay (for both general patients and behavioral health patients). We hypothesized that patients in emergency departments with superior nurse work environments would experience lower rates of leaving without being seen and shorter ED lengths of stay.

Methods

STUDY DESIGN/SETTING

This cross-sectional study used linked data from multiple sources including: the 2021 RN4CAST-New York/Illinois nurse survey,[19] the 2019 American Hospital Association (AHA) Annual Hospital Survey,[20] and the Center for Medicare and Medicaid Services' (CMS) publicly available 2021 Hospital Compare timely care data files.[21]

The RN4CAST-NY/IL is a unique survey of nurses as informants of the organizations in which they work. For this study, we used survey responses from registered nurses working in emergency departments to provide information on the nurse work environment. The RN4CAST-NY/IL survey data were collected from April to June 2021 from a 100% sample of registered nurses licensed in New York and Illinois.[22,23] Emailed surveys were sent through the

National Council of State Boards of Nursing and took approximately 10 to 15 minutes for nurses to complete. Nurses provided the name of their employer, which allowed us to aggregate responses from nurses working in the same hospitals and also to link the nurse survey data with other hospital data sources—in this case, to the CMS Hospital Compare files and AHA data.

Data collection from surveyed nurses followed a modified Dillman approach, including regularly scheduled follow-up reminders of non-respondents over a 1-month period. An overview of the RN4CAST-NY/IL methodology is detailed in prior literature.[12,22,24] The nurse-level response rate was 14%, which is within the usual range of response rates for electronic surveys.[12]

SAMPLE

The study sample consisted of emergency departments in non-federal acute care hospitals in New York and Illinois, including pediatric emergency departments. A minimum of 2 nurse respondents per hospital was identified as sufficient to produce reliable estimates of the nurse work environment, based on intraclass correlation coefficients.[25,26] The final sample included 1,189 nurses working in 215 emergency departments.

MEASUREMENTS

Exposure Variable

The primary exposure variable in this study was the nurse work environment derived from an abbreviated version of the Practice Environment Scale of the Nursing Work Index (PES-5)[27,28] in the RN4CAST-NY/IL survey. The original PES-NWI is a National Quality Forum-endorsed measure consisting of 31-items in 5 subscales which ask nurses to indicate their level of agreement that the following organizational attributes were present in their job: 1) nurse participation in hospital affairs; 2) nursing foundations for quality care; 3) nurse management ability; 4) staffing and resource adequacy; and 5) interprofessional collaboration.[27,28] The 31-item PES-NWI Cronbach alpha ranges from 0.84 to 0.93.[15]

The 5 items in the abbreviated PES-5 (Cronbach's alpha 0.82) are derived from each of the aforementioned 5 subscales of the full 31-item PES-NWI and demonstrate strong predictive validity compared to the full PES-NWI when used to evaluate the odds of 30-day patient mortality. The 5 items included in the PES-5 from the 31-item PES-NWI were chosen because they had the highest R-squared value in prior published work.[27,28] The PES-5 has been published previously as a valid tool to evaluate the nurse work environment.[13] Each individual emergency nurse response to the PES-5 was averaged to create a continuous value that was then aggregated to the hospital level. The aggregated hospital-level nurse work environment variable was then divided into quartiles, with the top quartile defined as emergency departments with better work environments, the bottom as poor, and the middle quartiles as mixed.[14,29,30] emergency departments within the bottom 25th percentile of the work environment were considered "poor," while "mixed" were in the 50th percentile, and "better" work environments were in the top 25th percentile.[31,32]

Outcome Variables

CMS' Hospital Compare data files provided the hospital-level study outcome measures. The percentage of patients who left the emergency department before being seen includes patients who were seen by the triage nurse but left the emergency department before being seen by a provider. Higher percentages indicate a higher proportion of patients leaving the waiting room before provider evaluation.[6,33,34] ED LOS is defined by the median time (in minutes) spent in the emergency department before discharge.[35,36]

The ED LOS was also separately defined by patients seen in the emergency department for behavioral health reasons on Hospital Compare.[37] Behavioral Health (BH) ED LOS was similarly defined by the median time (in minutes) that BH patients spent in the emergency departments before discharge. Unlike for general ED LOS, BH ED LOS also includes time spent awaiting transfer/transportation to home or outside facilities after their disposition was set to "discharged" since these patients often require 1:1 supervision. The BH population for this measure was identified through ICD-10 Principal Diagnosis codes (eg, mood disorders, anxiety disorders, intentional self-harm).

Covariates

The AHA Annual Survey provided data on hospital characteristics used as covariates in the models. Covariates included teaching status (non-teaching: no residents/fellows; minor teaching: <1 fellow/resident per 4 beds; major teaching: ≥1 fellow/resident to 4 beds), annual ED patient volume, and an indicator variable for whether or not the hospital was a designated trauma center.

ANALYSIS

Descriptive statistics (medians, standard deviations, range) were used to describe the distribution of timely ED outcomes across hospital characteristics (ie, teaching status)

TABLE 1
Emergency department timely care outcomes and characteristics in 215 study hospitals

Timely care outcome			Left without being seen (%)			ED length of stay (min)			BH ED length of stay (min)		
Hospital characteristics	**%**	**N**	**Median (SD)**	**Min-Max**	***P* value**	**Median (SD)**	**Min-Max**	***P* value**	**Median (SD)**	**Min-Max**	***P* value**
Overall		215	2.2 (2)	0-11		200.0 (50)	66-370		339.1 (127)	111-738	
Work environment					.02			<.001			.03
Poor	25.1	54	2.2 (2)	0-8		205.2 (40)	97-279		361.8 (137)	111-711	
Mixed	52.6	113	1.8 (2)	0-11		208.1 (52)	99-346		345.3 (131)	111-738	
Better	22.3	48	1.2 (1)	0-6		174.8 (49)	66-370		293.3 (89)	146-604	
ED annual volume					.04			<.001			<.001
0-20,000	14.4	31	1.0 (1)	0-5		142.1 (43)	66-275		241.7 (81)	111-395	
20-40,000	26.2	56	1.6 (2)	0-11		186.9 (39)	110-310		322.1 (111)	164-586	
40-60,000	17.2	37	2.1 (2)	0-6		218.5 (43)	162-330		359.7 (116)	186-669	
60-80,000	13.5	29	1.7 (1)	0-4		205.4 (52)	107-346		341.5 (141)	192-738	
80-100,000	8.4	18	2.3 (2)	0-8		227.2 (59)	136-370		397.2 (151)	223-658	
>100,000	20.5	44	2.0 (1)	0-6		226.1 (25)	170-276		370.4 (128)	162-625	
Teaching status					.003			<.001			<.001
No	38.1	82	1.3 (1)	0-6		173.6 (44)	66-279		301.2 (115)	111-669	
Minor	30.2	65	2.0 (2)	0-11		209.2 (49)	120-346		370.0 (126)	164-711	
Major	31.6	68	2.2 (2)	0-8		222.8 (43)	107-370		352.9 (133)	162-738	
Trauma center					.35			<.001			.08
No	47.4	102	1.6 (2)	0-11		186.0 (55)	66-346		321.3 (135)	111-711	
Yes	52.6	113	1.8 (2)	0-8		213.0 (42)	107-370		353.2 (120)	162-738	

BH, behavioral health patients; ED, emergency department; SD, standard deviation.
"Left without being seen" represents the percentage of patients who left the emergency department waiting room being seen by a provider. "ED Length of Stay" represents the average number of minutes patients spent in the emergency department before leaving the visit. "ED BH length of stay" represents the average number of minutes BH patients spent in the emergency department before being sent home. Study covariates: Emergency department volume, teaching status, and trauma center designation.

and poor, mixed, or better nurse work environments. To address skewness, the LWBS variable was log-transformed prior to analysis to improve the normality of the distribution for more precise statistical modeling.

Hospital-level linear regression models were constructed to estimate the association between the emergency nurse work environment and hospital-level ED outcomes, unadjusted and adjusted for hospital covariates. The model associations were constrained to be linear such that the effect of the nurse work environment on ED outcomes was the same for emergency nurse work environments that were poor vs mixed and mixed vs better. The linear regression coefficients estimating the effect of the work environment on LWBS were exponentiated and subtracted from 1 to indicate a percent change in LWBS, in methodological alignment with log-linear transformation procedures.[38,39]

We estimated the additional minutes that could be gained (ie, for patient care activities) associated with an emergency nurse work environment improvement (either from poor to mixed or from mixed to better) by multiplying the highest annual ED patient volume in each category of Table 3 by the difference in ED LOS (projected ED LOS minus baseline LOS) based on the beta coefficients from our linear regression models. The linear regression coefficients for LOS were multiplied by 2 that would be associated with improvements in the work environment from "poor" to "better" based on the linear constraint assumption. To compute the additional number of patients that could be seen if emergency departments improved their environments, we divided the estimated additional minutes associated with the new ED LOS using the same beta coefficients from our models.

TABLE 2
Unadjusted and adjusted regression coefficients estimating the effect of the emergency nurse work environment on timely emergency department outcomes in 215 study hospitals

Patient outcomes	Unadjusted model coefficient (95% CI)	Adjusted model coefficient (95% CI)
ED length of stay	−14.5 (−24.2, −4.9)*	−12.6 (−21.5, −3.8)*
BH ED length of stay	−33.4 (−59.3, −7.5)*	−33.1 (−58.9, −7.3)*
Left without being seen	0.79 (0.69, 0.89)†	0.81 (0.71, 0.93)‡

BH, behavioral health patients; CI, confidence interval; ED, emergency department.
"ED length of stay" represents the average number of minutes patients spent in the emergency department before leaving the visit. "ED BH Length of Stay" represents the average number of minutes BH patients spent in the emergency department before being sent home. "Left without being seen" represents the percentage of patients who left the emergency department waiting room being seen by a provider. Study covariates: Emergency department volume, teaching status, and trauma center designation. The hospital sample across all models ranged from 195 to 215 hospitals based on study outcome missingness across models. Coefficients for ED length of stay and BH ED length of stay represent reductions in minutes given a change in the work environment from poor to mixed or mixed to better as measured by the Practice Environment Scale of the Nursing Work Index. Coefficients for left without being seen represent a 21% (1-0.79) and 19% (1-0.81) reduction in the proportion of patients leaving the ED before being seen by a provider given a change in the work environment from poor to mixed or mixed to better.

* $P \leq .05$ level
† $P \leq .001$ level
‡ $P \leq .01$ level

StataIC 17 was used for data analyses. The University of Pennsylvania institutional review board approved the protocol for this study.

Results

ED outcomes and characteristics among the 215 emergency departments in the sample are described in Table 1. On average, emergency departments with the highest LWBS rates and longest ED lengths of stay (including BH) were large, major teaching trauma centers. Hospitals with an average ED annual volume of 80,000 to 100,000 patients experienced a median LWBS of 2.3%, a general ED LOS of 227.2 minutes (3.8 hours), and BH LOS of 397.2 minutes (6.7 hours). On average, emergency departments with poor work environments (as reported by nurses) had the worst median LWBS rates of 2.2% and BH LOS of 361.8 minutes (6 hours), while emergency departments with mixed environments had the longest overall patient LOS of 208.1 minutes (3.5 hours).

The results of the linear regression analyses are displayed in Table 2, with beta coefficients providing estimates of the association of the nurse work environment with ED outcomes unadjusted and adjusted for hospital covariates. The models for LOS can be interpreted as follows: after adjusting for hospital characteristics, an improvement in the nurse work environment from poor to mixed (or mixed to better) was associated with shorter patient LOS of 13 minutes (95% CI −21.5, −3.8, $P < .05$) and 33 minutes (95% CI −58.9, −7.3, $P < .05$) for patients with behavioral health needs. Improving the work environment from poor to mixed (and mixed to better) was associated with lower ED LWBS rates by a factor of 0.81 (95% CI 0.71, 0.93, $P \leq .001$) or a 19% reduction (1-0.81) in LWBS rates. A change in the work environment from poor to better is estimated to be associated with a 26-minute reduction in overall patient LOS (13 minutes × 2) and 66 minutes (33 minutes × 2) for patients with BH needs; and a 33% reduction in LWBS rates (0.81^2).

Estimates of expected annual additional patient-care minutes gained and number of patients seen are described in Table 3. We provide estimates for emergency departments of various patient volumes. Estimates indicate the projected annual number of patients that could be associated with a change in emergency nurse work environments from poor to better. On average, we project improvements in the emergency nurse work environment are associated with an additional 504,000 to 3,024,000 care minutes via a reduction in the general patient ED LOS. These estimates were even greater for patients presenting with BH care needs. Accounting for reductions in LOS for all patients, improvements in the emergency work environment are associated with an additional 11,824 to 41,071 patients seen annually in the emergency department, depending on annual patient volume. These projections represent an increase in ED throughput, wherein reductions in LOS may facilitate greater volumes of patients being evaluated for conditions in the emergency department in a timely

TABLE 3
Estimated patient care minutes gained and number of patients seen in the emergency department based on improving the emergency nurse work environment from poor to better

Timely care outcomes	ED length of stay (min)				BH ED length of stay (min)				Total patients
ED patient volume	Baseline ED LOS (minutes)	Improved ED LOS (minutes)	Additional care minutes	Additional patients seen	Baseline ED LOS (minutes)	Improved ED LOS (minutes)	Additional care minutes	Additional patients seen	Total additional patients
up to 20,000	142.1	116.9	504,000	4,311	241.7	175.7	1,320,000	7,513	11,824
40,000	186.9	161.7	1,008,000	6,234	322.1	256.1	2,640,000	10,308	16,542
60,000	218.5	193.3	1,512,000	7,822	359.7	293.7	3,960,000	13,483	21,305
80,000	205.4	180.2	2,016,000	11,188	341.5	275.5	5,280,000	19,165	30,353
100,000	227.2	202.0	2,520,000	12,475	397.2	331.2	6,600,000	19,928	32,403
120,000	226.1	200.9	3,024,000	15,052	370.4	304.4	7,920,000	26,018	41,071

BH, behavioral health patients; ED, emergency department.
"Left without being seen" represents the percentage of patients who left the emergency department waiting room being seen by a provider. "ED length of stay" represents the average number of minutes patients spent in the emergency department before leaving the visit. "ED BH length of stay" represents the average number of minutes BH patients spent in the emergency department before being sent home. Baseline estimates are derived from Table 1, while "Improved" estimates are derived by subtracting the coefficients from the linear regression models for ED length of stay (12.6 minutes) and BH ED length of stay (33.1 minutes). The linear regression coefficients are multiplied by 2 based on the linear constraint assumption indicating a change in the work environment from "poor" to "better." 'Additional care minutes' are calculated by multiplying the annual volumes by the change in length of stay times (baseline-improved length of stay time). 'Additional patients seen' are calculated by dividing additional care minutes by the improved length of stay minutes.

way that prevents clinical deterioration and other avoidable harm.[40,41]

Discussion

To our knowledge, this is the first study to evaluate the relationship between the emergency nurse work environment and patient outcomes, including the rates of patients who leave without being seen and ED lengths of stay. We found that more favorable emergency nurse work environments were associated with lower LWBS rates and shorter ED LOS. We estimate that a change in the emergency nurse work environment (from poor to better) could contribute to more patients being seen in the emergency department due to additional patient care minutes attributed to shorter patient ED LOS. For example, we project that hospitals with the greatest annual patient volumes (>100,000) could gain an additional 3,024,000 patient care minutes, which translates into seeing approximately 15,052 more general ED patients, and for patients presenting with BH needs, 7,920,000 additional patient care minutes which translated to roughly 26,018 more BH patients annually. Combining the reductions in LOS among all patients, improvements in the emergency nurse work environment could be associated with an additional 11,824 to 41,071 patients seen annually, with estimates varying depending on the emergency departments' annual patient volume. Our findings suggest that investing in the nurse work environment may be associated with increases in ED throughput, which is a key priority for Healthy People 2030[1,42] and health care organizations across the U.S.

Minutes of additional time is valuable specifically in the ED context—this time can be used to triage more patients, evaluate a waiting room patient, transport a boarded patient from the emergency department to an inpatient bed, or stabilize a patient arriving in a trauma bay. Our estimates of additional expected patient care minutes and numbers of patients seen demonstrate the projected benefits associated with superior emergency nurse work environments through increased throughput. Increased throughput, measured by reductions in ED patient LOS and LWBS are not only associated with better outcomes for timely conditions like acute myocardial infarction and stroke,[7,43] but are also associated with increased patient satisfaction.[44] Additional care minutes in the ED setting are a crucial resource given that the unexpected *is expected* at all times in this care setting.[45,46] Given that many hospitals promote quality initiatives and performance evaluations in emergency departments to reduce LWBS and patient LOS, the findings from this study are highly relevant and informative to hospital administrators and managers seeking solutions to improve timely ED care.

This study provides important insight into the understudied relationship between nurses' work environments

and timely care in the emergency department. The nurse work environment represents the foundation from which nurses deliver their care by measuring nurses' perceived autonomy, leadership support, staffing/resources, and interprofessional collaboration in the ED setting.[27] Our findings suggest that nurses' unfavorable work environments are associated with prolonged ED lengths of stay and higher rates of patients leaving without evaluation by a provider. These results extend existing evidence demonstrating a relationship between poor emergency nurse work environments and suboptimal patient care quality reports by emergency nurses.[13]

Our study findings emphasize the importance of health care administrator investments in high-quality emergency nurse work environments at the entryway to acute care services. Recent evidence demonstrates that emergency nurses are more likely to rate their work environment as fair or poor compared to inpatient nurses (43.5% vs 36.5%, $P < .001$).[11,12] That approximately 1 in 5 emergency nurses (19.1%)[11] are not confident that their manager will act to resolve patient-related problems on their unit is concerning in the context of the present study where nurses manage the care of patients with prolonged ED lengths of stay and high LWBS rates. Given the nationwide hospital crowding crisis impacting emergency departments and the clinicians who care for them, hospital administrators should consider investing in strong manager-nurse relationships in emergency departments and nurse representation in initiatives aimed at improving timely outcomes such as LWBS rates and ED LOS. The CEO[47] of a large U.S. health care system's recent public support for policies that provide safer workloads for nurses is an example of measures that establish trusting relationships between frontline clinicians and health care decision-makers in the context of acute care access barriers that exacerbate nurse workload, such as ED crowding. Indeed, inadequate nurse staffing is addressed by key emergency medicine leaders as a modifiable feature of hospital work environments[48] that may mitigate the poor aspects of ED crowding.

Further, an evidence-based intervention endorsed by both nurses and physicians to improve their work environments is safer nurse staffing levels.[49] In the years just prior to the pandemic (including the time of the present study), nurses' work environments were characterized by high patient-to-nurse staffing ratios as high as 9:1 in New York City hospitals.[19] While California remains the only state to implement nurse staffing minimums (including a recommended emergency department staffing ratio of 1 nurse to every 4 patients),[50] nurse staffing legislation has been increasingly introduced across multiple states in the U.S., including New York, given the well-established relationship between poor nurse staffing and suboptimal patient outcomes (ie, hospital readmissions, prolonged hospital stays). Providing nurses with adequate nurse staffing on their units, through policy interventions such as safe nurse staffing mandates,[51] is an important step that hospital administrators should consider advocating for and implementing to provide nurses with safe working conditions for their patients.

Limitations

There are limitations in this study that warrant discussion. First, the sample was limited to 2 states and thus our findings do not nationally represent care outcomes for generalizability purposes. Despite this limitation, New York and Illinois include a large number of hospitals that are heterogeneous with respect to ED volumes, teaching status, nurse work environments, and trauma designation. Second, due to limitations in the data reported by Hospital Compare, this study did not account for inpatient hospital-level factors influencing timely ED care such as hospital boarding/crowding or the health outcomes of boarded patients. This is an important area for future research. While the AHA Annual Hospital Survey available for analysis was published in 2019 and merged with 2021 nurse survey and Hospital Compare data, we expected that any differences in hospital characteristics (eg, technology capability) between these years would not significantly impact study results. Finally, our nurse response rate should be contextualized within the primary purpose of this study which focused on ED-level factors of the work environment on ED-level timely care outcomes. Germaine to our study purpose was not the response rate of individual nurses, but rather the extent to which the nurse responses provide reliable hospital-level estimates of nurse work environments in a large, representative sample of emergency departments.

Implications for Emergency Nurses

Our findings suggest that the nurse work environment is an important feature of emergency departments associated with variations in timely care outcomes measured by LWBS rates and ED LOS. Emergency nurses are the first contact point for patients entering the emergency department; thus, the context of their care—the work environment—holds important implications for throughput issues of patients leaving without being seen and experiencing prolonged ED lengths of stay. Investing in nurses' care environment may serve as an important solution for these suboptimal patient outcomes.

Conclusion

Lack of timely care in emergency departments is a patient access and quality and safety concern. Nurses are directly embedded throughout all aspects of ED timely care, and our study demonstrates that their work environments are associated with higher rates of patients who leave without being seen and longer ED lengths of stay. Investments in better nurse work environments characterized by adequate nurse staffing, supportive nurse leadership, nurses' involvement in unit affairs, and strong interprofessional relations should be considered by hospital administrators as features that may contribute to improved patient throughput in the context of hospital crowding.

Author Disclosures

Conflicts of interest: none to report.

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